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Wildfire Spread Prediction Using Attention Mechanisms in U2-NET

Hongtao Xiao · Yingfang Zhu · Yurong Sun · Gui Zhang · Zhiwei Gong

Environmental Science, Computer Science · Forests · 27 September 2024

TLDR A neural network model based on the U2-Net architecture is trained and introduced both inside and outside the nested U-shaped structure to enable the model to suppress the activation of irrelevant areas. Expand

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FirePred: A hybrid multi-temporal convolutional neural network model for wildfire spread prediction

Mohammad Marjani · Seyed Ali Ahmadi · M. Mahdianpari

Environmental Science, Computer Science · Ecological Informatics · 1 September 2023

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Incorporating Stochastic Wind Vectors in Wildfire Spread Prediction

Sahar Masoudian · Jason Sharples · Z. Jovanoski · I. Towers · S. Watt · Environmental Science · Atmosphere · 27 October 2023

The stochastic nature of environmental factors that govern the behavior of fire, such as wind and fuel, exposes **wildfire** modeling to a degree of uncertainty. In order to produce more realistic… Expand

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Wildfire Spread Prediction Using Attention Mechanisms in U-NET

Kamen Shah · Maria Pantoja · Environmental Science, Computer Science · 2023 3rd International Conference on Electrical… · 19 July 2023

TLDR Predicting **wildfire spread** using NN with an attention mechanism to improve spatial recognition and overall performance in neural networks for **wildfire spread prediction** and uses ensemble modeling to reduce bias and variation, leading to more consistent and accurate **predictions**. Expand

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Real Time Monitoring of Forest Fires and Wildfire Spread Prediction

Aayush Subramaniam · Adithya S Rao · Akshat Vajrapu · Arya Darshan Roi V. · Vanamala H.R

Environmental Science, Computer Science · 2023 2nd International Conference on Applied… · 4 May 2023

TLDR Together, real-time monitoring and burnt area **prediction** can provide valuable information and support for effective forest fire management and response. Expand

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Assessing WildfireGPT: a comparative analysis of AI models for quantitative wildfire spread prediction

M. Ramesh · Ziheng Sun · +4 authors · Daniel Tong · Environmental Science, Computer Science · Natural Hazards · 28 May 2025

TLDR This study examines the application of **Wildfire**GPT for **wildfire** forecasting, focusing on its limitations in quantitative predicting Fire Radiative Power (FRP) **spread** and comparing its performance with a specialized predictive model based on TabNet, emphasizing the necessity of domain-specific models for accurate and actionable **wildfire** forecasting. Expand

4 PubMed Save Cite

WFNet: A hierarchical convolutional neural network for wildfire spread prediction

Wenyu Jiang Yuming Qiao +6 authors Fei Wang Environmental Science, Computer Science · Environmental Modelling & Software · 1 September 2023

9 Publisher Save Cite

A Probabilistic Approach to Wildfire Spread Prediction Using a Denoising Diffusion Surrogate Model

Wenbo Yu Anirbit Ghosh T. S. Finn Rossella Arcucci M. Bocquet Sibo Cheng

Environmental Science, Computer Science · arXiv.org · 1 July 2025

TLDR This study presents the first denoising diffusion model for predicting **wildfire spread**, a new kind of AI framework that learns to simulate fires not just as one fixed outcome, but as a range of possible scenarios. Expand

1 PDF arXiv Save Cite

Assessment of deep learning models integrated with weather and environmental variables for wildfire spread prediction and a case study of the 2023 Maui fires

Jiyeon Kim Yingjie Hu Negar Elhami-Khorasani Kai Sun Ryan Zhenqi Zhou

Environmental Science, Computer Science · 5 September 2025

TLDR This work assesses the ability of five typical deep learning models integrated with weather and environmental variables for **wildfire spread prediction** based on over ten years of **wildfire** data in the state of Hawaii and identifies important weather and environmental factors associated with the 2023 Maui **wildfires**. Expand

[PDF] arXiv Save Cite

Vi-Net: A Hybrid Semantic Segmentation Approach for Enhanced Wildfire Spread Prediction

Manavjit Singh Dhindsa Sagarika Naik +4 authors T. Ravichandran

Environmental Science, Computer Science · Annual International Computer Software and... · 8 July 2025

TLDR Experimental results show that Vi-Net achieves an F1-score of ~97% and an Intersection over Union (IoU) of ~94% on test data, significantly outperforming standalone U-Net and ViT models, underscore Vi-Net's potential to improve **wildfire** mitigation planning, resource allocation, and emergency response. Expand

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